number 6.However, in the sixth period after lanthanum, the electrons are preferentially filled in inner 4f sub shell and these fourteen elements following lanthanum show similar chemical properties. Therefore these elements are grouped together and placed at the bottom of the periodic table. This position can be justified as follows.

1. Lanthanoids have general electronic configuration [Xe] $4f^{1-14}$ $5d^{0-1}$ $6s^2$
2. The common oxidation state of lanthanoides is +3
3. All these elements have similar physical and chemical properties.

Similarly the fourteen elements following actinium resemble in their physical and chemical properties. If we place these elements after Lanthanum in the periodic table below 4d series, the properties of the elements belongs to a group would be different and it would affect the proper structure of the periodic table. Hence a separate position is provided to the inner transition elements as shown in the figure.

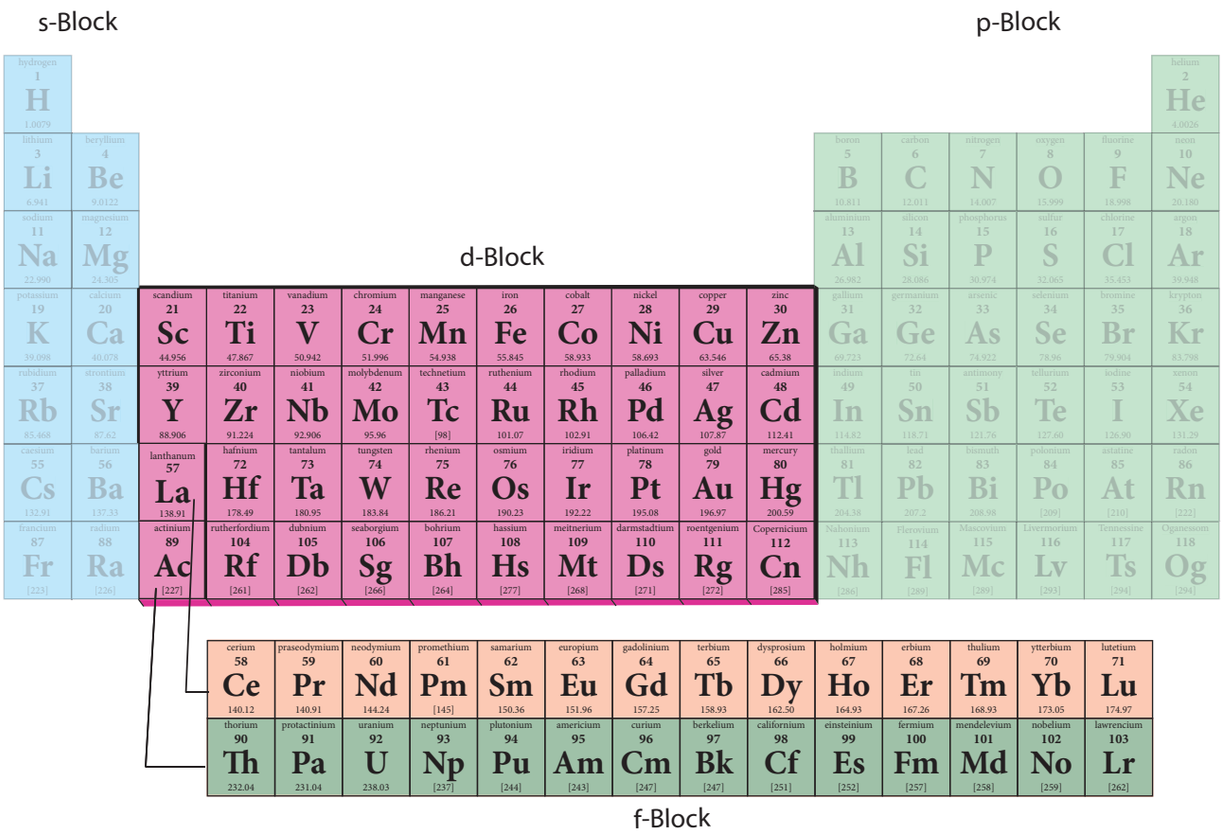


Figure 4.10 position of inner transition elements

Electronic configuration of Lanthanoids:

We know that the electrons are filled in different orbitals in the order of their increasing energy in accordance with Aufbau principle. As per this rule after filling 5s,5p and 6s and 4f level begin to fill from lanthanum, and hence the expected electronic configuration of Lanthanum(La) is [Xe] $4f^1$ $5d^0$ $6s^2$ but the actual electronic configuration of Lanthanum is

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